



RUBY ON RAILS CHEAT SHEET

Core setup, initialization, operations & integration quick reference

PART 1: SETUP & CORE

01. SETUP & CLI

Installation

```
gem install rails
rails new myapp --api --database=postgresql
rails server (or rails s)
rails generate scaffold User name:string email:string
rails db:migrate
rails console (REPL)
```

04. CONTROLLERS

Workflow

```
class UsersController < ApplicationController
  before_action :authenticate_user!
  before_action :set_user, only: [:show, :update]
  def index
    @users = User.all
    render json: @users
  end
  def create
    @user = User.new(user_params)
    if @user.save
      render json: @user, status: :created
    else
      render json: @user.errors, status: :unprocessable_entity
    end
  end
end
```

02. MVC STRUCTURE

Architecture

app/models/	ActiveRecord models
app/controllers/	ApplicationController subclasses
app/views/	ERB templates
app/helpers/	view helper methods
config/routes.rb	routing config
db/migrate/	migration files

05. ACTIVE RECORD MODELS

CRUD API

```
class User < ApplicationRecord
  validates :email, presence: true, uniqueness: true
  validates :name, length: { minimum: 2 }
  has_many :orders, dependent: :destroy
  belongs_to :team, optional: true
  before_save :downcase_email
end
```

03. ROUTES

YAML / Config

```
resources :users # CRUD routes
resources :posts, only: [:index, :show, :create]
namespace :api do
  resources :users
end
get '/login', to: 'sessions#new', as: :login
root 'home#index'
rails routes # list all routes
```

06. MIGRATIONS

Integration

```
rails generate migration AddPhoneToUsers phone:string
class AddPhoneToUsers < ActiveRecord::Migration
  def change
    add_column :users, :phone, :string
    add_index :users, :email, unique: true
  end
end
rails db:migrate / db:rollback / db:schema:dump
```



SECURITY, MONITORING & OPERATIONS

Production hardening, observability, performance tuning & CLI reference

PART 2: OPS & SECURITY

07. VIEWS (ERB)

Hardening

```
<%= @user.name %> <%= comment %>
<% @users.each do |u| %><%= u.name %><% end %>
<%= render 'shared/header' %>
<%= link_to 'Edit', edit_user_path(@user) %>
<%= form_with model: @user do |f| %>
  <%= f.text_field :name, class: 'form-control'...
<% end %>
```

10. QUERY INTERFACE

Unit Tests

```
User.all / User.find(1) / User.find_by(email:...
User.where('active = ?', true).order(:name).l...
User.where(active: true, role: 'admin')
User.joins(:orders).where(orders: {status: 'p...
User.select(:id, :name).distinct
User.count / User.sum(:balance) / User.averag...
```

08. ACTIVE RECORD ASSOCIATIONS

Observability

```
has_one has_many belongs_to has_and_belong...
has_many :comments, through: :posts # join m...
# Eager loading:
User.includes(:orders, :role).all
User.eager_load(:orders).where(orders: {statu...
User.preload(:orders)
```

11. RAILS GENERATORS

Commands

```
rails g controller Api::Users index show crea...
rails g model Post title:string body:text use...
rails g migration CreateJoinTableUsersRoles u...
rails g job SendEmail
rails g mailer UserMailer welcome_email
bundle exec rails test (or rspec with rspec-...
```

09. VALIDATIONS

Optimization

```
validates :name, presence: true
validates :email, format: { with: URI::MailTo...
validates :age, numericality: { greater_than:...
validates :role, inclusion: { in: %w[admin us...
validate :custom_validation
def custom_validation
  errors.add(:base, 'Custom error') if condition
end
```

12. COMMON GOTCHAS

Warning

N+1: use includes/eager_load never loop + lazy load
Strong parameters: use permit() to whitelist params
SECRET_KEY_BASE in credentials use rails credentials:edit
Time zones: always use Time.current, not Time.now
Convention over configuration follow Rails naming conventions